PERCABANGAN, PERULANGAN, DAN OPERASI GIT LANJUTAN PADA DART

LATIHAN PERTEMUAN 3

PEMROGRAMAN MOBILE

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Class : 2G

Major : Business Information System

Github : https://github.com/Lovie-Tonimba/semester4-PemrogramanMobile-LatihanPertemuan3.git

Studi Kasus: Aplikasi Kalkulator BMI

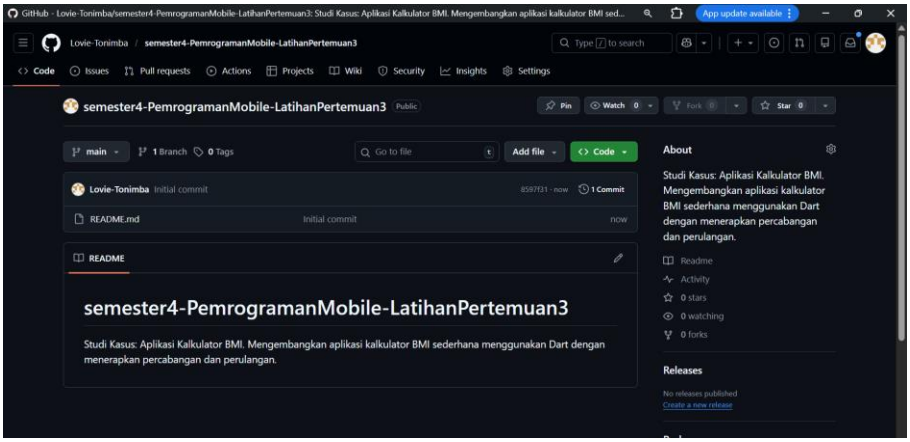
Kita akan mengembangkan aplikasi kalkulator BMI sederhana menggunakan Dart dengan menerapkan percabangan dan perulangan.

Fitur Aplikasi:

- Input tinggi dan berat badan
- Kalkulasi BMI
- Kategorisasi hasil (kurus, normal, gemuk)
- Riwayat perhitungan (menggunakan perulangan)

Alur Kerja Git:

1. Buat repository baru di GitHub



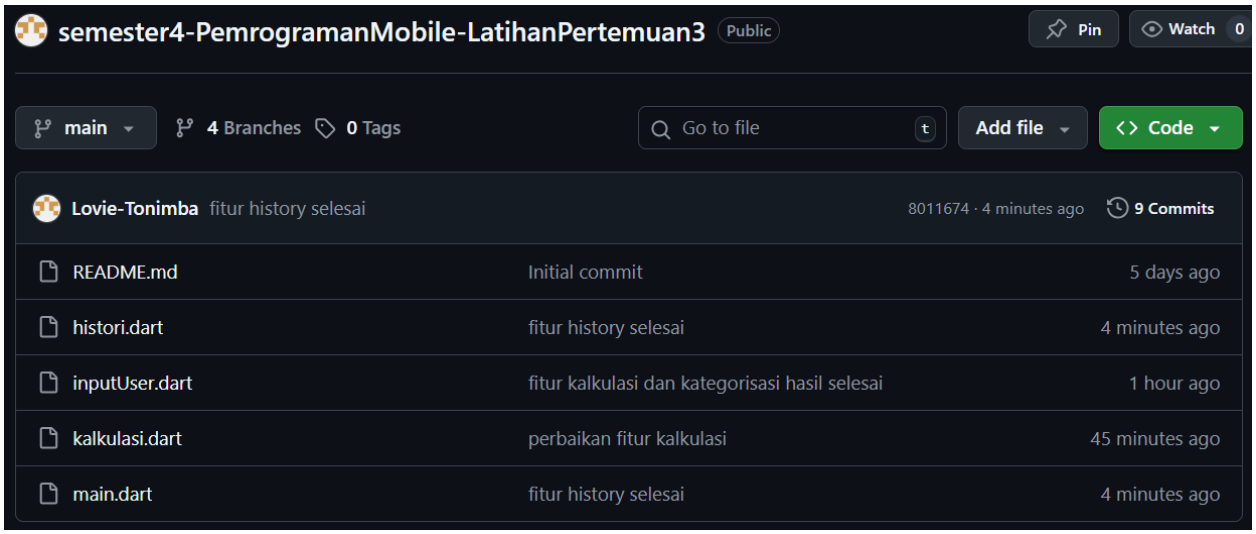
2. Clone repository ke local

```
PS C:\Users\lovie tonimba> git clone https://github.com/Lovie-Tonimba/semester4-PemrogramanMobile-LatihanPertemuan3.git
Cloning into 'semester4-PemrogramanMobile-LatihanPertemuan3'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
PS C:\Users\lovie tonimba>
```

3. Buat branch untuk setiap fitur:
    - Branch input-validation
    - Branch bmi-calculation
    - Branch history-feature

```
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git branch input-validation
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git branch bmi-calculation
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git branch history-feature
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3>
```

4. Merge branch ketika fitur selesai
5. Push ke repository remote



Branch input-validation

```
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git pull origin main
From https://github.com/Lovie-Tonimba/semester4-PemrogramanMobile-LatihanPertemuan3
 * branch            main       -> FETCH_HEAD
Updating 8597f31..6659095
Fast-forward
 main.dart | 6 ++++++
 1 file changed, 6 insertions(+)
 create mode 100644 main.dart
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3>
```

```dart
import 'dart:io';

Map<String, double> inputBMI() {
  double berat = 0, tinggi = 0;

  while (true) {
    stdout.write("Masukkan berat (kg) : ");
    double inputB = double.tryParse(stdin.readLineSync() ?? "") ?? 0;
    stdout.write("Masukkan tinggi (cm) : ");
    double inputT = double.tryParse(stdin.readLineSync() ?? "") ?? 0;

    if (inputB <= 0 || inputT <= 0) {
      print('Input harus berupa angka valid dan lebih dari 0');
      print("----------------------------------------------");
      continue;
    } else {
      return {'berat': inputB, 'tinggi': inputT};
    }
  }
}
```

```
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git merge input-validation
Updating 6659095..bf764c4
Fast-forward
 inputUser.dart | 23 +++++++++++++++++++++++
 main.dart      |  3 +++
 2 files changed, 26 insertions(+)
 create mode 100644 inputUser.dart
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Lovie-Tonimba/semester4-PemrogramanMobile-LatihanPertemuan3.git
   6659095..bf764c4  main -> main
```

Branch bmi-calculation

```
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git checkout bmi-calculation
Switched to branch 'bmi-calculation'
Your branch is up to date with 'origin/bmi-calculation'.
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git pull origin main
From https://github.com/Lovie-Tonimba/semester4-PemrogramanMobile-LatihanPertemuan3
 * branch            main       -> FETCH_HEAD
Updating bf764c4..ce16103
Fast-forward
 inputUser.dart | 7 ++-----
 1 file changed, 2 insertions(+), 5 deletions(-)
```

```dart
import 'inputUser.dart';

List<String> riwayatBmi = [];

void kalkulasi() {
  Map<String, double> data = inputBMI();

  double berat = data['berat']!;
  double tinggi = data['tinggi']!;

  // ubah tinggi dari cm ke meter dulu
  double tinggiMeter = tinggi / 100;
  // rumus BMI yaitu berat / (tinggi * tinggi)
  double hasilBmi = berat / (tinggiMeter * tinggiMeter);

  String status;
  if (hasilBmi < 18.5) {
    status = "Kurus";
  } else if (hasilBmi >= 18.5 && hasilBmi <= 24.9) {
    status = "Ideal/Normal";
  } else {
    status = "Gemuk";
  }

  print('Berat Badan  : $berat kg');
  print('Tinggi Badan : $tinggi cm');
  print('Skor BMI     : $hasilBmi');
  print('Status       : $status');

  // hasil masuk ke dalam list riwayat secara otomatis
  riwayatBmi.add("$berat, $tinggi, $status");
  print('Data berhasil masuk ke riwayat bmi');
}
```

```
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git merge bmi-calculation
Updating ce16103..309c2d3
Fast-forward
 inputUser.dart |  4 +---
 kalkulasi.dart | 27 ++++++++++++++++++++++++++
 main.dart      |  6 +++---
 3 files changed, 31 insertions(+), 6 deletions(-)
 create mode 100644 kalkulasi.dart
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Lovie-Tonimba/semester4-PemrogramanMobile-LatihanPertemuan3.git
   ce16103..309c2d3  main -> main
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3>
```

Branch history-feature

```
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git checkout history-feature
Switched to branch 'history-feature'
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git pull origin main
From https://github.com/Lovie-Tonimba/semester4-PemrogramanMobile-LatihanPertemuan3
 * branch            main       -> FETCH_HEAD
Updating 8597f31..309c2d3
Fast-forward
 inputUser.dart | 18 ++++++++++++++++++
 kalkulasi.dart | 27 +++++++++++++++++++++++++++
 main.dart      |  9 +++++++++
 3 files changed, 54 insertions(+)
 create mode 100644 inputUser.dart
 create mode 100644 kalkulasi.dart
 create mode 100644 main.dart
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3>
```

```dart
import 'dart:io';
import 'kalkulasi.dart';

void histori() {
  while (true) {
    stdout.write("Apakah ingin melihat riwayat perhitungan? ");
    String? input = stdin.readLineSync()?.toLowerCase();

    if (input == null || input.isEmpty) {
      print('Inputan tidak boleh kosong silahkan pilih ya atau tidak');
      continue;
    }

    if (input == 'ya') {
      print("-- Daftar Riwayat Perhitungan BMI --");
      if (riwayatBmi.isEmpty) {
        print('Belum ada data.');
        break;
      } else {
        for (int i = 0; i < riwayatBmi.length; i++) {
          print('${i + 1} = ${riwayatBmi[i]}');
        }
      }
    } else if (input == 'tidak') {
      print('Terimakasih sudah menggunakan sistem ini');
      break;
    }

    stdout.write("\nApakah ingin menghitung BMI lagi? (ya/tidak): ");
    String? inputLagi = stdin.readLineSync()?.toLowerCase();
    if (inputLagi == 'ya') {
      kalkulasi();
    } else if (inputLagi == 'tidak') {
      print('Terimakasih sudah menggunakan sistem ini');
      break;
    } else {
      print('Invalid input, try again');
    }
  }
}
```

```
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git merge history-feature
Updating 0913a97..8011674
Fast-forward
 histori.dart | 40 ++++++++++++++++++++++++++++++++++++++++
 main.dart    |  2 ++
 2 files changed, 42 insertions(+)
 create mode 100644 histori.dart
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Lovie-Tonimba/semester4-PemrogramanMobile-LatihanPertemuan3.git
   0913a97..8011674  main -> main
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3>
```

Class main

```dart
import 'histori.dart';
import 'kalkulasi.dart';

void main() {
  print("===== APLIKASI KALKULATOR BMI =====");
  // input user
  // bmi kalkulasi dan kategori hasil
  kalkulasi();
  // bmi history perhitungan
  histori();
}
```

Output

```
===== APLIKASI KALKULATOR BMI =====
Masukkan berat (kg) : 55
Masukkan tinggi (cm) : 155
Berat Badan  : 55.0 kg
Tinggi Badan : 155.0 cm
Skor BMI     : 22.892819979188342
Status       : Ideal/Normal
Data berhasil masuk ke riwayat bmi
Apakah ingin melihat riwayat perhitungan? ya
-- Daftar Riwayat Perhitungan BMI --
1 = 55.0, 155.0, Ideal/Normal
```

```
Apakah ingin menghitung BMI lagi? (ya/tidak): ya
Masukkan berat (kg) : 60
Masukkan tinggi (cm) : 170
Berat Badan  : 60.0 kg
Tinggi Badan : 170.0 cm
Skor BMI     : 20.761245674740486
Status       : Ideal/Normal
Data berhasil masuk ke riwayat bmi
Apakah ingin melihat riwayat perhitungan? ya
-- Daftar Riwayat Perhitungan BMI --
1 = 55.0, 155.0, Ideal/Normal
2 = 60.0, 170.0, Ideal/Normal

Apakah ingin menghitung BMI lagi? (ya/tidak): tidak
Terimakasih sudah menggunakan sistem ini
PS C:\Users\lovie tonimba\semester4-PemrogramanMobile-LatihanPertemuan3>
```